Limit State Design of Singly & Doubly Reinforced RCC Beams

IS 456 code-compliant structural design manual covering neutral axis depth, moment of resistance, shear stirrup spacing, and crack control.

CampusVerse Academic Library • Civil Engineering > Structural Engineering > Reinforced Concrete Design (RCC)

---

1. Limit State Design Principles (IS 456:2000)

Ensures safety against bending and shear collapse and serviceability deflection limits (span/250). Limiting neutral axis Xu,max / d is 0.48 for Fe 415 steel.

2. Under-Reinforced Rectangular Sections

Under-reinforced sections ensure tension steel yields prior to concrete crushing, providing ductile warning before ultimate failure occurs.